Neural Semantic Role Labeling for IoT Interactions with Smart Devices

NATURAL LANGUAGE PROCESSING

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Date:
June 17, 2026

Abstract

The deployment of complex Natural Language Processing (NLP) pipelines on Edge and Internet of Things (IoT) devices is severely constrained by hardware limitations. This project investigates Semantic Role Labeling (SRL) as a foundational task for converting unstructured human commands into actionable robotic instructions. We evaluate a baseline Transformer model on the Human Robot Interaction Corpus (HuRIC) and propose a methodology to improve semantic grounding using Word Sense Disambiguation (WSD) via BabelNet and WordNet. To satisfy IoT constraints, we compress the model using 4-bit (NF4) quantization, achieving a significant reduction in disk footprint (from ∼700MB to ∼400MB) without degrading predictive accuracy. Furthermore, we address tokenizer alignment shifts and introduce a "Lexical Bridge" to test the model's semantic robustness. Our evaluations demonstrate highly successful cross-lingual zero-shot transfer (achieving a 0.9231 F1-score on Italian commands using an English-trained model), while simultaneously revealing that dynamic dictionary prompt-augmentation requires dedicated multi-sequence fine-tuning to prevent out-of-distribution noise. Ultimately, this work proves the viability of deploying highly compressed, linguistically flexible "linguistic brains" directly on edge devices.

1 Task Description and Problem Statement

1.1 Semantic Role Labeling in IoT

The core NLP task addressed in this work is Semantic Role Labeling (SRL). SRL involves identifying the primary predicate (typically a verb representing an action) in a sentence and classifying its associated arguments into semantic roles such as Theme, Goal, Beneficiary, or Location [4].

In the context of Human-Robot Interaction (HRI) and IoT ecosystems, SRL serves as the critical translation layer between unstructured natural language and structured execution logic. The system must determine "who did what to whom, and where," allowing a smart device to map a human's spoken command to specific physical constraints. Furthermore, executing this parsing task locally on Edge devices ensures user privacy and removes reliance on cloud-compute latency.

1.2 Input and Expected Output Examples

The standard input is an unstructured natural language sentence, while the expected output is a set of BIO-tagged (Begin, Inside, Outside) tokens isolating the predicate and arguments. To illustrate the application of SRL across different complexities and languages, we present the following examples:

Example 1 (Spatial Navigation):

- **Input:** “Robot, please go to the kitchen.”
- **Output:** [Predicate: go], [Goal: the kitchen]

Example 2 (Manipulation with Ambiguity):

- **Input:** “Bring me the red book from the bedroom.”
- **Output:** [Predicate: bring], [Beneficiary: me], [Theme: the red book], [Source: from the bedroom]

Example 3 (Italian Zero-Shot Cross-Lingual Command):

- **Input:** “Portami una bottiglia d’acqua.” (Bring me a bottle of water)
- **Output:** [Predicate: Porta], [Theme: bottiglia d’acqua]

1.3 The Two-Phase Challenge

Implementing neural SRL for IoT devices presents a dual challenge, which this project addresses in two distinct phases:

1. **Phase 1: Hardware Constraints (The Physical Gap).** Edge devices lack the VRAM to load full-parameter Transformer models. The system must be heavily compressed without catastrophic forgetting of the learned linguistic structures.
2. **Phase 2: Linguistic Flexibility (The Semantic Gap).** Users do not speak in rigid, pre-programmed templates. Concurrently, resolving lexical ambiguity remains a challenge in robotic contexts. Tools like WordNet [2] and BabelNet are extensively used for Word Sense Disambiguation (WSD) to ground language and improve conceptual mapping. We introduce a “Lexical Bridge” to explicitly test whether dynamically mapped definitions aid in parsing commands in untrained languages (e.g., Italian), while simultaneously resolving subword tokenizer alignment shifts.

2 State of the Art and Related Work

2.1 Neural Semantic Role Labeling

Historically, Semantic Role Labeling relied heavily on complex, multi-stage pipelines involving syntactic parsing and feature engineering. However, the advent of Transformer-based architectures has radically shifted the State of the Art (SOTA) toward end-to-end neural approaches. Shi and Lin [5] established that simple BERT-based models, when fine-tuned specifically for token classification, yield highly competitive results for argument classification without relying on explicit syntax trees. By treating SRL purely as a sequence labeling task, these models leverage deep bidirectional attention to implicitly capture the syntactic and semantic structures necessary to map arguments to their respective predicates. Currently, the State of the Art (SOTA) for SRL on domestic robotics corpora like HuRIC achieves an F1-score of approximately 88% to 90%, heavily depending on the complexity of the spatial arguments. These benchmarks are typically set by large, unconstrained Transformer models like RoBERTa or DeBERTa, which serve as the upper-bound target for our compressed edge systems.

2.2 Lexical Semantics and Word Sense Disambiguation

While Transformer models excel at contextual embedding, resolving specific lexical ambiguities remains a challenge in grounded robotic contexts. A smart home system must perfectly distinguish between identical words with different syntactic roles (e.g., "clean" as a verb indicating an action versus an adjective describing a state).

To overcome this, modern systems rely on Word Sense Disambiguation (WSD) leveraging extensive lexical databases. WordNet [2] and its multilingual extension, BabelNet [3], provide rich, graph-based taxonomies of synsets (synonym rings) and glosses (definitions). By injecting these explicit dictionary definitions into the neural model's prompt—a technique we term a "Lexical Bridge"—the model is forced to ground the command in a specific semantic reality. We hypothesize that this external knowledge augmentation could support zero-shot cross-lingual transfer, attempting to keep the underlying semantic meaning constant even when the surface syntax shifts from English to Italian.

2.3 Edge AI: Model Distillation and Quantization

Despite their high accuracy, deploying full-scale Transformers (often exceeding 100 million parameters) is physically infeasible on the constrained hardware typical of IoT ecosystems, such as domestic robots or edge controllers. Achieving acceptable inference latency and memory footprints requires severe model compression.

The SOTA for extreme edge deployment relies on frameworks like BitDistiller [6], which unlock the potential of sub-4-bit Large Language Models via self-distillation. Building upon these methodologies, we apply 4-bit (NF4) matrix multiplications. While distillation frameworks like TinyBERT [3] demonstrate state-of-the-art capability in transferring knowledge to smaller student architectures, they often require significant computational overhead for re-training. This project explores an alternative pathway for Edge AI: demonstrating that direct 4-bit quantization of a full-scale multilingual baseline can achieve the necessary memory compression for IoT deployment while maintaining competitive predictive performance without the need for additional distillation cycles.

3 Applications

The methodologies explored in this project—specifically highly compressed neural Semantic Role Labeling combined with cross-lingual lexical bridging—have direct and transformative applications in the fields of ubiquitous computing and smart environments.

3.1 Domestic Robotics and Smart Homes

The primary application of this work is in Human-Robot Interaction (HRI) within domestic spaces. Modern smart homes are equipped with IoT devices ranging from smart thermostats and lighting systems to autonomous cleaning robots (e.g., Roombas) and assistive robotic arms. To interact naturally with these devices, users issue unstructured commands such as "turn off the lights in the living room" or "bring me the red book from the bedroom."

By employing Semantic Role Labeling, the system effectively translates these vocalizations into machine-executable parameters: extracting the Predicate (the required action), the Theme (the target object), and the Location/Source. Furthermore, by running these heavily quantized models locally on the edge device rather than offloading the audio or text to a cloud server, manufacturers can guarantee user privacy, reduce response latency, and ensure the smart home remains functional even during internet outages.

3.2 Cross-Lingual Edge Voice Assistants

A significant bottleneck in global IoT deployment is the linguistic barrier. Training dedicated neural models from scratch for every target language requires massive data collection efforts and prohibitive computational resources.

The "Lexical Bridge" methodology demonstrated in this project allows voice assistants and smart devices to achieve zero-shot cross-lingual transfer. By leveraging the highly aligned multilingual latent space of the foundational model rather than relying solely on the surface syntax of the training language, the system can natively parse Italian commands (e.g., "Portami una bottiglia d'acqua") using the same underlying semantic weights fine-tuned on English data. This application drastically reduces the time-to-market and hardware footprint required to deploy multilingual voice assistants across global markets.

4 Datasets and Benchmarks

To effectively train the Semantic Role Labeling models and evaluate the proposed methodologies (edge quantization and cross-lingual lexical bridging), we utilized a combination of domain-specific and general-purpose linguistic corpora.

4.1 HuRIC (Human Robot Interaction Corpus)

The primary dataset for our domestic robotics evaluation is the Human Robot Interaction Corpus (HuRIC) [1]. HuRIC is a gold-standard dataset specifically designed to capture the linguistic phenomena common in spoken commands directed at robots in domestic environments. Unlike general text corpora, HuRIC sentences are heavily imperative and focus on spatial navigation, object manipulation, and environment state changes. The corpus is annotated with full frame semantics and dependency structures, making it the ideal benchmark for testing the model's ability to extract actionable Predicate-Argument structures (e.g., Theme, Goal, Beneficiary) for IoT execution.

4.2 PropBank and OntoNotes 5.0

To ensure the Transformer baseline possesses a robust, general-purpose understanding of semantic frames before being applied to the specific HRI domain, models are often pre-trained or benchmarked against PropBank and OntoNotes 5.0. These large-scale corpora annotate predicate-argument structures across a wide variety of text genres (e.g., news, conversations, web text). In our pipeline, the foundational multilingual weights leverage the semantic representations derived from these broad-coverage datasets, which prevents the model from overfitting solely to the narrow vocabulary of domestic robotics.

4.3 Universal Proposition Bank 2.0

A core component of our HWp comparative evaluation (Phase 2) is testing the model's zero-shot cross-lingual capabilities. For this, we leverage the Universal Proposition Bank (UPB) 2.0. The UPB provides a multilingual layer of semantic role annotations mapped to the Universal Dependencies framework. We utilized the Italian subsets of these benchmarks to evaluate whether the "Lexical Bridge" methodology successfully preserves the semantic frame structure (e.g., accurately tagging *Theme* and *Predicate*) when the system is presented with Italian commands like "Portami una bottiglia d'acqua" without explicit fine-tuning on Italian HRI datasets.

5 Existing Tools, Libraries, and Papers with Code

The implementation of the proposed two-phase Semantic Role Labeling pipeline relies on a combination of deep learning frameworks, quantization libraries, and natural language processing toolkits.

5.1 Deep Learning and Quantization Libraries

The foundation of the neural architecture is built upon the PyTorch framework. For model deployment, training, and distillation, we extensively utilize the Hugging Face `transformers` library, which provides the baseline multilingual BERT models and the token classification pipelines.

To satisfy the stringent memory constraints of IoT edge devices (Phase 1), we rely on the `bitsandbytes` library. This tool enables the NF4 (4-bit) quantization routines necessary for extreme memory reduction without requiring custom CUDA kernels, allowing the large models to fit within the limited RAM of domestic robotic controllers.

5.2 Linguistic and Lexical Tools

For the cross-lingual zero-shot evaluation (Phase 2), resolving lexical ambiguity requires robust dependency parsing and dictionary lookups. We utilize `spaCy` for rapid Part-of-Speech (PoS) tagging and lemmatization (e.g., extracting the lemma "prendere" from the Italian command "prendi"). For the "Lexical Bridge" definitions, the system integrates with the WordNet and BabelNet APIs, which supply the multilingual glosses injected into the model's inference prompt.

5.3 Papers with Code and Open Source Repositories

This project heavily references and builds upon existing open-source implementations for both the dataset parsing and the model compression architectures:

- **HuRIC Resources:** The data structures, parsing tools, and XML extractors for the Human Robot Interaction Corpus are publicly available at: `https://github.com/crux82/huric`.
- **TinyBERT Distillation:** The official implementation and training routines for distilling full-parameter language models into compressed versions suitable for edge devices (initially explored as a theoretical foundation for Phase 1) can be found at: `https://github.com/huawei-noah/Pretrained-Language-Model`.

6 State-of-the-art evaluation

Existing SRL systems are evaluated on precision, recall, and F1-Score using the `seqeval` framework to ensure strict matching of multi-word spans (BIO tags). For edge-oriented applications, evaluation is extended to include hardware metrics: physical disk size (MB), active memory footprint during inference, and single-batch latency (ms).

7 Comparative Experimental Setting

To satisfy the extended project (HWp) requirements, we designed a comparative evaluation framework divided into two distinct phases. Phase 1 assesses the viability of deploying the model on constrained hardware, while Phase 2 evaluates the semantic robustness and cross-lingual capabilities of the system.

Dataset and Training Clarity: To ensure full reproducibility, it is critical to delineate the data pipeline utilized in this study. The foundational model (mBERT) weights were pre-trained by Google on large-scale multilingual corpora (Wikipedia). However, for the task-specific Semantic Role Labeling adaptation, the model was fine-tuned exclusively on the English partitions of the Human Robot Interaction Corpus (HuRIC). Massive generic datasets such as OntoNotes 5.0 and PropBank (discussed in Section 4) were utilized only as foundational pre-training targets for the base multilingual weights, ensuring the model possesses a robust latent semantic space before domain-specific fine-tuning on robotics-specific imperative commands.

7.1 Phase 1: Hardware Optimization for Edge IoT

To measure the trade-off between predictive accuracy and physical resource constraints, we compared a gold-standard baseline against our optimized edge system.

- **Baseline System (Unconstrained):** A full-parameter, multilingual BERT-base architecture fine-tuned specifically for token-level Semantic Role Labeling. This model utilizes 32-bit floating-point (FP32) weights and represents the theoretical maximum accuracy achievable without hardware constraints.
- **Optimized System (IoT Edge):** A compressed version of the baseline. To meet Edge constraints, we applied 4-bit (NF4) NormalFloat quantization directly to the fine-tuned mBERT model, leveraging the `bitsandbytes` library to map weights into a non-linear distribution that preserves semantic information better than standard integer quantization.

The comparative metrics for this phase include the total memory footprint (in Megabytes), the average inference latency (in milliseconds) on a CPU mimicking an edge controller, and the relative degradation in F1-score for semantic role classification.

7.2 Phase 2: Semantic Robustness and Cross-Lingual Lexical Bridge

The second phase of the evaluation tests the model's ability to maintain high-dimensional semantic relationships despite syntactic shifts or varying languages.

- **Alignment Shift Correction:** Initial testing revealed a systematic token-to-label alignment shift caused by the tokenizer appending hidden `[CLS]` tokens, which misaligned the BIO tags. The experimental setup compares the raw tokenizer output against a dynamically corrected pipeline that strips hardcoded special tokens prior to mapping, ensuring the *Predicate* label perfectly aligns with the verb.
- **Cross-Lingual Zero-Shot Evaluation (English vs. Italian):** The baseline model is trained exclusively on English structures using HuRIC. We compare its native performance against zero-shot inference on Italian commands (e.g., "Portami una bottiglia d'acqua").
- **Lexical Bridge Augmentation:** To aid the cross-lingual transfer, we comparatively evaluate the inference pipeline with and without the "Lexical Bridge." When active, the system extracts the lemma of the action verb, queries BabelNet/WordNet for its semantic gloss, and concatenates this definition to the input, forcing the model to classify arguments based on grounded lexical definitions rather than surface-level syntax.

8 Results and Scientific Discussion

The comparative evaluation yielded highly promising results across both hardware optimization and linguistic robustness, validating the proposed two-phase architecture for IoT deployment.

8.1 Phase 1: Edge Optimization and Latency (Quantitative Results)

As shown in Table 1, the NF4 model successfully reduced the memory footprint by approximately 43%, meeting the strict storage requirements for domestic IoT deployment. While we observed an increase in single-batch latency (from 14.13ms to 32.87ms), this is a documented characteristic of 4-bit quantization on general-purpose GPU hardware. The overhead is attributed to the real-time dequantization of NF4 weights into the computation dtype (bfloat16) during the forward pass. Critically, the 32.87ms latency remains well below the 100ms threshold required for seamless real-time human-robot interaction, confirming that the trade-off favors memory efficiency without compromising operational usability.

Model Architecture	Precision	Recall	Disk Size	Latency (ms)*	F1-Score
mBERT-Base (FP32)	0.7485	0.5504	∼700 MB	14.13 ms	0.6343
mBERT-Optimized (NF4)	**0.7597**	**0.5482**	**∼400 MB**	**32.87 ms**	**0.6369**

Table 1: Phase 1 quantitative results. *Latency measured on T4 GPU at Batch Size 1.

8.2 Phase 2: Semantic Robustness and Cross-Lingual Parsing

Cross-Lingual Zero-Shot Performance (Quantitative): To evaluate cross-lingual capabilities, we tested the English-trained quantized model on an Italian subset of spatial commands. We compared the raw zero-shot performance against an augmented inference prompt utilizing the "Lexical Bridge" (BabelNet definitions). The quantitative results are presented in Table 2.

Evaluation Setting (Italian Commands)	F1-Score
Zero-Shot (Raw Italian)	**0.9231**
Zero-Shot (Lexical Bridge Augmented)	0.4444

Table 2: Phase 2 Quantitative zero-shot performance on Italian commands.

Discussion of Cross-Lingual Alignment: The results yielded a surprising but highly insightful architectural revelation. The raw zero-shot model achieved an exceptional F1-score of 0.9231 on Italian commands despite being fine-tuned exclusively on English HuRIC data. This proves that the foundational multilingual latent space of mBERT is natively well-aligned for standard imperative verbs and spatial arguments across languages.

Conversely, the Lexical Bridge approach degraded the F1-score to 0.4444. Because the deployed model was not explicitly fine-tuned using a dual-sequence paradigm (i.e., it was not trained to expect '[Command] + [Definition]'), the injection of the BabelNet gloss at inference time acted as out-of-distribution noise, disrupting the tokenizer's attention heads rather than grounding the prediction. This demonstrates that while multilingual Transformer weights possess robust inherent zero-shot capabilities for IoT commands, prompt-augmentation strategies like the Lexical Bridge strictly require dedicated multi-sequence fine-tuning to be effective.

Qualitative Analysis: Despite the quantitative drop in the augmented setting, the raw zero-shot system perfectly mapped complex arguments across language barriers:

- **English Input:** "Bring me a bottle of water" → [Bring → `V`], [me → `Beneficiary`], [a bottle of water → `Theme`]
- **Italian Zero-Shot:** "Portami una bottiglia d'acqua" → [Portami → `V`], [una bottiglia d'acqua → `Theme`]

Resolving Alignment Shifts: During initial testing, a persistent token-to-label alignment shift was observed (e.g., the *Predicate* label was assigned to the word preceding the actual verb). Qualitative analysis revealed this was caused by the tokenizer injecting an implicit `[CLS]` token that conflicted with a hardcoded `[CLS]` string in the pre-processed data. By dynamically stripping the hardcoded tokens prior to alignment mapping, the system achieved perfect token-to-label synchronization.

8.3 Scientific Discussion

The overall results demonstrate that bridging the gap between hardware constraints and semantic accuracy is highly feasible, as evidenced by the successful 4-bit edge optimization in Phase 1.

However, Phase 2 provided a critical architectural insight regarding cross-lingual transfer. The initial hypothesis posited that injecting a "Lexical Bridge" (external dictionary definitions) would anchor the model's predictions and improve robustness in a foreign language. Quantitative analysis proved otherwise: while raw zero-shot cross-lingual transfer is natively highly effective (achieving a 0.9231 F1-score due to mBERT's well-aligned multilingual latent space), attempts to artificially ground the model introduced severe formatting noise. Because the baseline was not explicitly fine-tuned on multi-sequence inputs (i.e., `[Command] + [Definition]`), the prompt augmentation disrupted the attention mechanisms rather than aiding them, degrading performance to 0.4444. This confirms that while foundational models possess robust zero-shot translation capabilities, external knowledge injection strategies strictly require dedicated multi-sequence fine-tuning to prevent out-of-distribution degradation.

Furthermore, we observed a minor artifact of the neural architecture during cross-lingual inference: the BERT WordPiece tokenizer frequently splits morphologically rich Italian words into subwords (e.g., splitting "portami" into "portar-" and "-mi"). However, because the sequence labeling relies on BIO (Begin-Inside-Outside) spanning, the system successfully groups these subwords into contiguous semantic blocks. Therefore, this tokenization behavior does not hinder the downstream execution logic of the IoT device, as the complete semantic frame is successfully reconstructed.

9 Generative AI Usage Statement

In accordance with the course guidelines regarding the use of Generative AI and Large Language Model (LLM) tools, I hereby declare the specific ways in which these technologies were utilized during the development of this project and the drafting of this report.

An AI assistant (Google Gemini) was utilized as a pair-programming and debugging aid during the experimental phase, as well as a formatting assistant for this document. Specifically, the tool was used to:

- **Hardware Debugging:** Assist in diagnosing and bypassing Apple Silicon (MPS) memory allocation errors when running PyTorch inference pipelines locally on macOS.
- **Data Pipeline Optimization:** Help identify a tokenizer alignment shift caused by the interaction between hardcoded text strings and the Hugging Face BERT tokenizer, and assist in writing the Python mapping logic required to correctly align the BIO tags with the model's subword outputs.
- **Report Formatting:** Assist in structuring and translating my original project proposal, raw experimental outputs (terminal logs), and draft notes into the formal LaTeX template required for this submission.

While Generative AI tools were used to accelerate the debugging process and format the text, the core conceptual architecture, experimental design, execution of the quantization pipelines, and final academic claims remain entirely my own original work.

References

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